

PalmSource®

Conference and Expo



Advanced UI

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Anatomy of the Stock Event Loop

```
EventType event;

do
   {
      EvtGetEvent(&event, evtWaitForever);
      if (! SysHandleEvent(&event))
         if (! MenuHandleEvent(0, &event, &error))
            if (! AppHandleEvent(&event))
               FrmDispatchEvent(&event);
   }
while (event.eType != appStopEvent);
```



Event Loop

- FrmDispatchEvent passes the event FIRST to the handler that was registered (usually by AppHandleEvent)
- If the event wasn't handled (i.e. the handler returns false) it flows to FrmHandleEvent, where it receives default handling
  - Keystrokes flow into fields
  - FrmUpdateEvent calls FrmDrawForm()



Opaque UI Structures

Myth:

UI structures are documented so you can edit their contents.



Opaque UI Structures

Reality:

UI structures are documented to help debugging.

(And that’s it.)



Usable and Visible

Myth:

Applications can control which form objects draw by setting the “visible” bit.



Usable and Visible

Reality:

The “visible” bit is actually state information, not something you should be setting.

- Use FrmShowObject() and FrmHideObject()



Clipboard

Myth:

The clipboard is just for boring text.



Clipboard

Reality:

You can put bitmaps on the clipboard, too!



Adding a Bitmap to the Clipboard

```
ClipboardAddItem(clipboardBitmap,
  bitmapP, MemPtrSize(bitmapP));
```

- Text and bitmap clipboards are separate; one doesn't overwrite the other



Custom Fonts

Myth:

Custom fonts aren’t supported.



Custom Fonts

Reality:

Custom fonts ARE supported.



Custom Fonts

Honest answer:

Custom fonts ARE supported, but we don’t make it easy (yet).



Creating a Custom Font

- Fonts are NFNT resources (well, almost)
  - Slight differences in header
  - Glyphs contain space
- No support in Constructor for creating fonts
- But there are various third-party tools for creating and importing fonts
  - xFont
  - PilRC
  - ResEdit (Macintosh)

Adding a Font to your Project (PC)

- Use xFont (freeware) to make a font in PilRC format. Call it something like “myFont.pfn”
  - Or create font directly in PilRC format
    See PilRC Manual for details
- Install PilRC CodeWarrior plugin
  http://www.calliopeinc.com/pilrcplugin.html
- Add a .rcp file to project with this line in it:
  - Font ‘FONT’ FONTID 128 “myFont.pfn”
- Compile and get a .r file containing a ‘NFNT’ resource
  - data 'NFNT' (20052) means Resource ID is 20052



Installing a Custom Font

```
MemHandle fontH;
FontType* fontP;

fontH = DmGetResource(fontResource, 25002);
fontP = (FontType*) MemHandleLock(fontH );
FntDefineFont(fntAppFontCustomBase, fontP);
MemHandleUnlock(fontH);
DmReleaseResource(fontH);
```

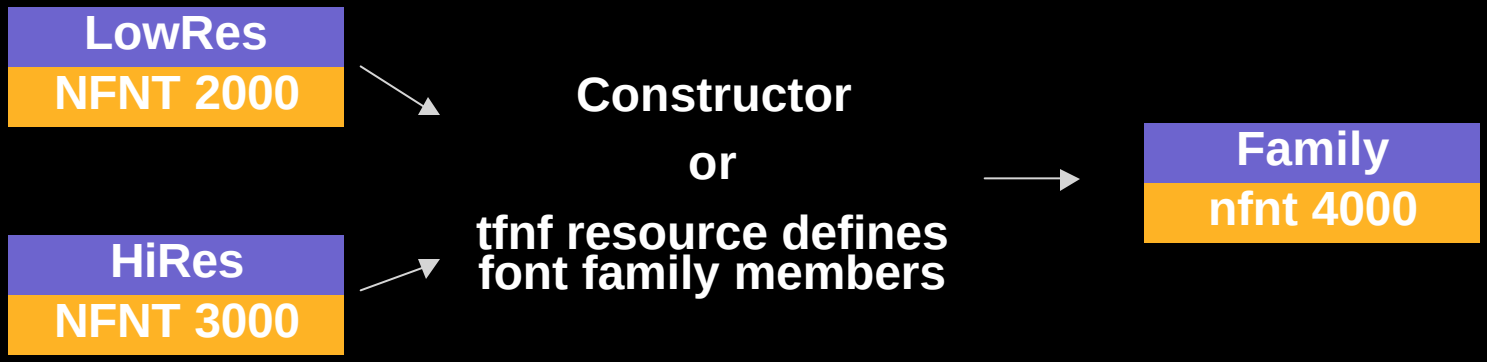
- **`fntAppFontCustomBase`** is the first ID available
  - add an offset when defining multiple fonts



Using a Custom Font

- WinDrawChars will now use this font
  - if you call **`FntSetFont(fntAppFontCustomBase)`**
- LstGlueSetFont, FrmGlueSetLabelFont, CtlGlueSetFont, TblSetItemFont can then take **`fntAppFontCustomBase`** to set the list, form, control or table font to the new custom font

High Resolution Font Families



- Use only on Palm OS 5 and later
  - Not backwards compatible (check OS version!)
- DmGetResource('nfnt', myFontFamilyID);
- From there use same as "old style" custom font



Tables

Myth:

Tables are Complicated



Tables

Reality:

Tables are Very Complicated



Do You Really Need a Table?

- Tables are ideal when:
  - You need UI widgets embedded in a table
    - fields, checkboxes, or anything that accepts input
- Consider alternatives when:
  - You simply want to display data in a multi-column format
  - You require scrolling
    - Tables have no inherent scrolling functionality!



Alternatives to Tables

- Lists
  - Can have multiple columns when drawn by callback
  - Great when rows can select together
- Fields + Scrollbar
  - Great for text-only data
- Gadgets
  - Most flexible, most work
  - Get a rectangle that receives hits
  - Implement only the functionality you need



Attention Manager

Myth:

The Attention Manager is a 4.0 replacement for the Alarm Manager.



Attention Manager

Reality:

The Attention Manager is a central UI repository for attention getting messages from all applications.

Step 1: Call the Attention Manager

- Often in response to an Alarm or Notification

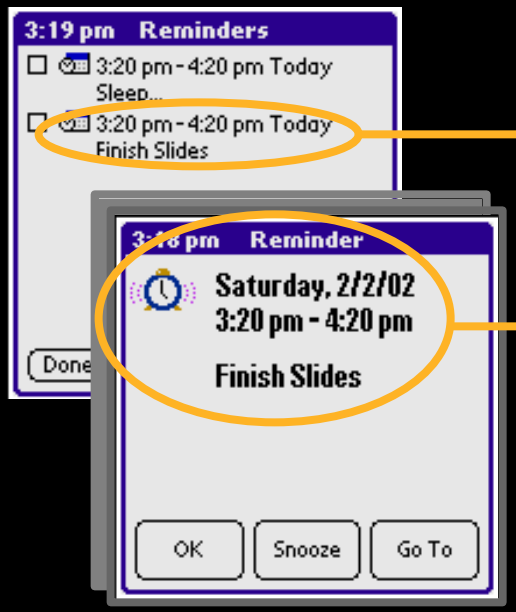
```
AttnGetAttention(cardNo, dbID, userData, NULL,
   kAttnLevelSubtle, kAttnFlagsUseUserSettings, 5, 2);
```



Handle this launch code:
`sysAppLaunchCmdAttention`

```
case sysAppLaunchCmdAttention:
       HandleAttention((AttnLaunchCodeArgsType*)cmdPBP);
```

Step 2: Draw the Attention UI




```
switch (cmdPBP->command)
```

```
case kAttnCommandDrawList:
```

```
case kAttnCommandDrawDetail:
```

```
x = cmdPBP->commandArgsP->drawList.bounds.topLeft.x;
y = cmdPBP->commandArgsP->drawList.bounds.topLeft.y;
WinDrawBitmap(iconP, x, y);
x += kAttnListTextOffset;
WinDrawChars(theStr, StrLen(theStr), x, y);
```



Step 3: Extras

- **`case kAttnCommandPlaySound`**
  - Chance to play a custom sound
  - Sent if requested in AttnGetAttention() call

- **`case kAttnCommandCustomEffect`**
  - Chance to do any other custom effect

Step 4: Handle User's Choice




```
case kAttnCommandGoThere:
    AttnForgetIt(cardNo, dbID, cmdPBP ->userData);
    buf = (UInt32*) MemPtrNew(sizeof(UInt32));
    MemPtrSetOwner(buf, 0);
    *buf = paramP->userData;

    SysUIAppSwitch(cardNo, dbID,
        sysAppLaunchCmdCustomBase, buf);
```

kAttnCommandSnooze

kAttnCommandGotIt

Handle **sysAppLaunchCmdCustomBase** in PalmMain()



Step 5: Update the Attention Manager

- **`AttnForgetIt`** – when something in the attention manager is no longer relevant
- **`AttnIterate`** – Iterates through everything that belongs to your app and gives your callback a chance to handle each one
  - Update all attentions
  - Erase all attentions
- **`AttnUpdate`** – Update specific attention
- **`AttnGetCounts`** – Use to find out how many attentions you have pending



Session Q&A



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